Longitudinal and time-series documentation protocols at the ADA - the development of a data dictionary

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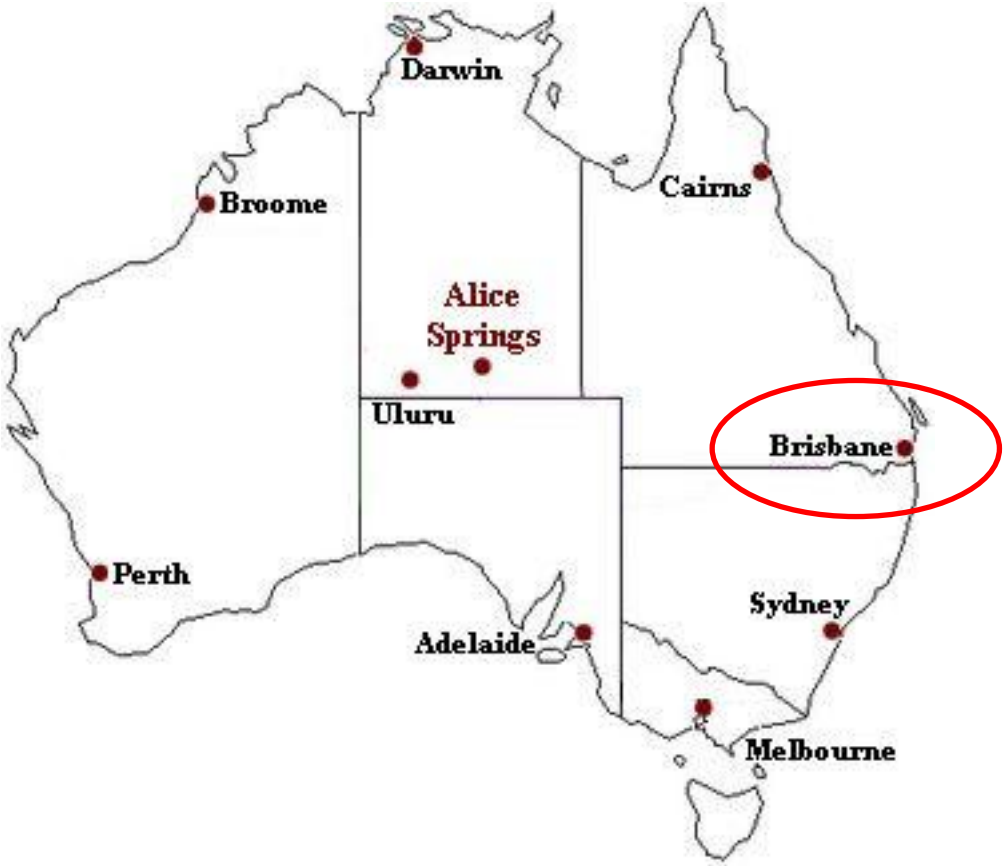
Longitudinal data sub-archive

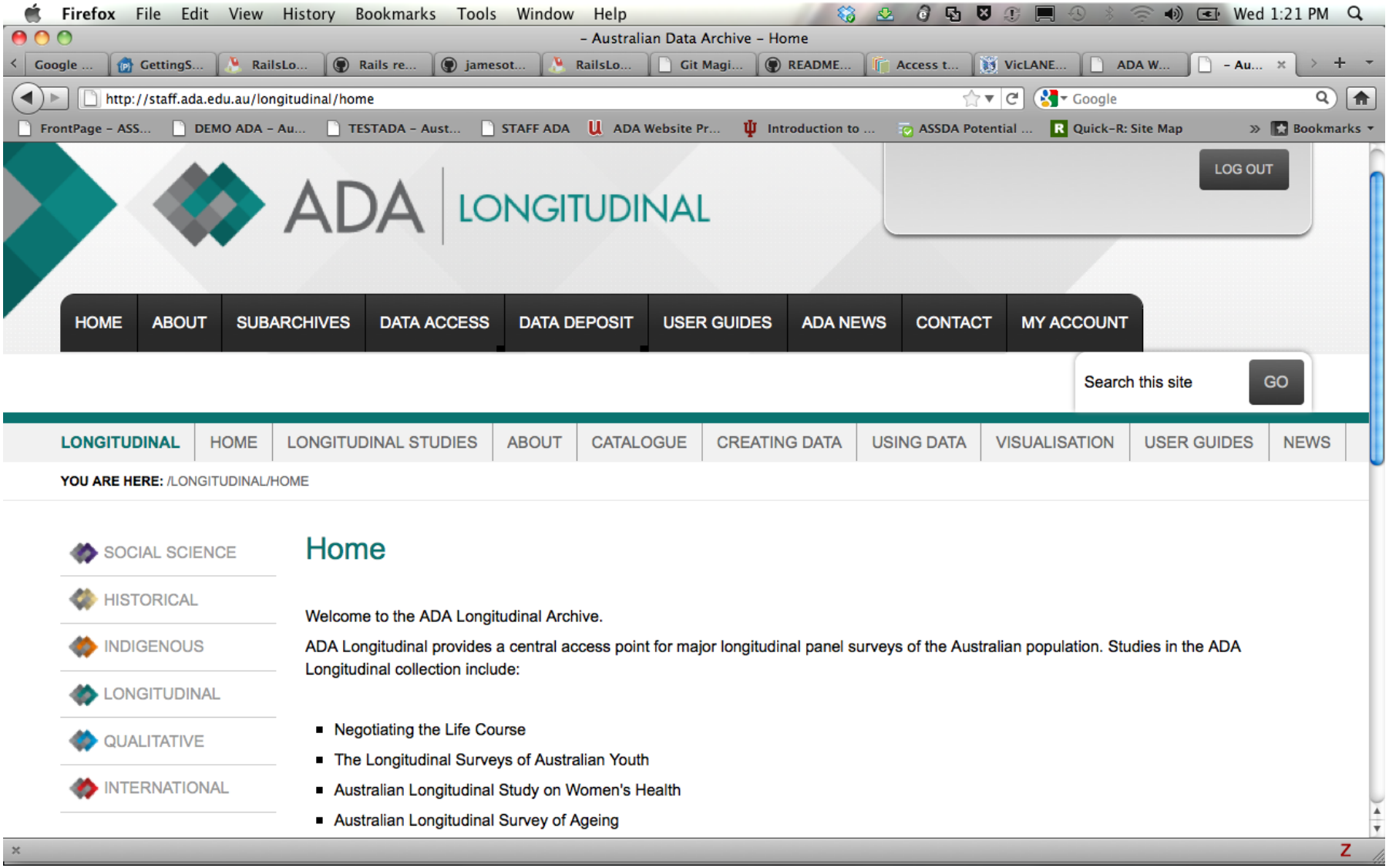


ASSDA or ADA?

- From July 2011, ASSDA will be changing names to ADA - The Australian Data Archive, with a new website:
    - http://www.ada.edu.au

ADA AUSTRALIAN DATA ARCHIVE

The future: ADA Longitudinal



Longitudinal data sub-archive

Repeated cross-sectional studies

> new sample at different points in time
> represents snapshot of population at each time point
> aspect of individual's change not available

Cohort studies

> group of individuals at a similar state in the life course, studied over time
> problems with drop-outs

Household panels

> Household as a study unit
> Number of individuals can vary (move in, move out)

Secondary Analysis

Negotiating the Life Course
started 1997, data collection every three years, four waves archived at ASSDA

designed to study the changing life courses and decision-making processes of Australian men and women as the family and society move from male breadwinner orientation in the direction of higher levels of gender equity.

Change in marital status for an individual across waves:
Are you married or in a relationship?

Wave	Variable name	Variable label
W1	Q20	Partneredstatus
W2	Q27	Q27 partnered status
W3	Q27	Q27 partnered status
W4	Q66	Are you married or in a relationship?

What is needed?

A so called data dictionary / data map

Question Text	Wave 1	Wave 2	Wave 3	Wave 4
Are you married or in a relationship?	Q20	Q27	Q27	Q2

Challenges of building data dictionaries:

- time consuming to prepare
- usually not enough funds or staff available
- ongoing work, needs updating after each wave

How can we facilitate the process?

Automate the process!

Variable format:

w1(Q29), w2(Q27), w3(Q27), w4(Q2)
Not in a relationship
In a relationship, not living together
Living in a relationship, but not legally married
Married and living with spouse

Literal Question:
Which of the following describes your current relationship?

Additional metadata preparation is required.

Getting the meta data

Meta data needed:

xml document

- Variable name
- Question Text

data set in STATA

- Variable name
- Variable label
- Value labels(responses)

- Variable name
- Question Text
- Variable label
- Value labels (responses)

A data set of meta data needed for each wave in STATA

Why draw variable and value labels from Stata? Depends on level of processing of waves, as to extent of metadata (particularly value labels)

Parse the xml document

```
- <var ID="V20" name="Q10b" files="F1" dcml="2" intrvl="contin">
    <location width="8" />
    <labl>SC Individual Civic Participation - public meeting</labl>
  - <qstn>
    - <qstnLit>
        <![CDATA[ During the last 12 months, without being paid, have
        you:

        Attended a public meeting  ]]>
      </qstnLit>
    </qstn>
```

- Using Python's xml.dom.minidom library
- parsing the xml document and extracting the required information
- saving it as a csv file, which then can be imported into STATA

```
mapping = {}
 for node in doc.getElementsByTagName("var"):
  name = node.getAttribute("name")
  mapping[name] = ""
  L = node.getElementsByTagName("qstnLit")
  for node2 in L:
    qstnLit= ""
    for node3 in node2.childNodes:
      if node3.nodeType == Node.TEXT_NODE:
        qstnLit += node3.data
      elif node3.nodeType == Node.CDATA_SECTION_NODE:
        qstnLit += node3.data

    mapping[name] = clean_question(qstnLit)
```

Get more meta data using STATA

name	type	isnumeric	format	vallab	varlab
q18a	byte	1	%8.0g	LABA	Ecometric SC Civic participation - petitio
Q7a	byte	1	%8.0g	LABA	CE skipping schoo
q18c	byte	1	%8.0g	LABA	Ecometric SC Civic Participation - joined to solve proble
Q12a	byte	1	%8.0g	LABB	SC Place attachmen
Q8c	byte	1	%8.0g	LABB	CE trust community member
q12j	byte	1	%8.0g	LABB	SC Intergenerational closure - adult role model

	vallab	value	label
13	LABA	-2	Very Unlikely
14	LABA	-1	Unlikely
15	LABA	0	Don't Know
16	LABA	1	Likely
17	LABA	2	Very Likely
18	LABA	99	Refused

Variable name	Value label	Value label detail
q18a	LABA	-2, very unlikely; -1, unlikely; 0, don't know; 1, likely; 2, very likely; 99, refused

Work in progress

• ‘linking variables’ program to be written in STATA (currently considering trailing blanks, upper/lower cases)

• “Same but Different” labels (eg. 1=‘yes’ 2=‘no’ versus 1=‘yes’ 0=‘no’)

• Variables that have no value labels assigned (age) might need an additional characteristic to be included into the comparison, such as the type of each variable (ie categorical/continuous/ordinal)

•variables without question text, such as derived variables and scales will need a substitute for the question text

• other challenges to be discovered (!)

Work in progress

•preliminary version of the data map to be sent to the depositor, covering identical and similar variables

• changes provided by the depositor will need to be incorporated into the latest version of the program

• the data map will be the reference for an online variable search tool

- search for a particular variable within a wave
- search within the question text for key words

Question Text	Wave 1	Wave 2	Wave 3	Wave 4
Are you married or in a relationship?	Q20	Q27	Q27	Q2

Further Plans

- click to download longitudinal data in either long or wide format, including selection of variables

- online data analysis

Thank you...

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